California Housing Price Prediction using Linear Regression

Objective

Predict house prices using the California Housing Dataset and evaluate a Linear Regression model.

Dataset Overview

Property	**Value**
Dataset	California Housing
Rows	20,640
Features	8
Target Variable	MedHouseVal

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1. Exploratory Data Analysis (EDA)

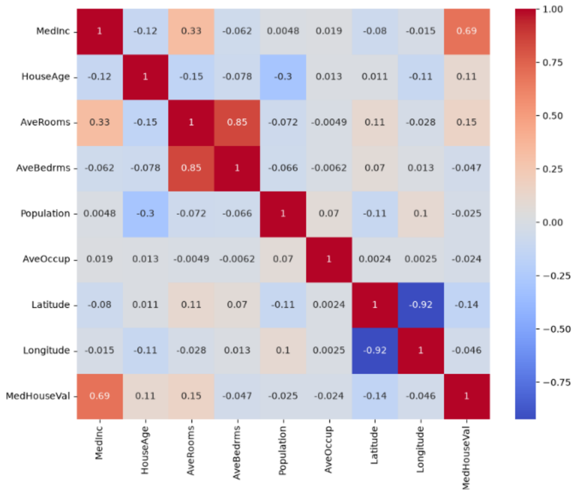
Data Quality

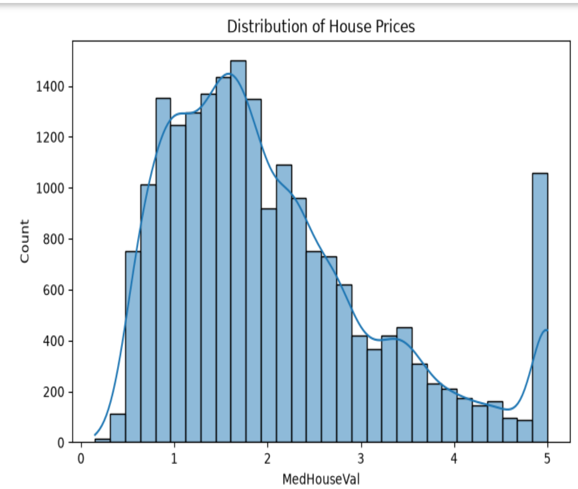
- No missing values found.
- All features are numerical.
- No categorical encoding was required.

Feature Analysis

- Median Income (MedInc) has a strong positive relationship with house prices.
- Latitude and Longitude also significantly affect house value.
- Population and Occupancy have weaker relationships.

Visualizations (Correlation Heatmap and House Price Distribution Plot)





2. Model Building

Data Preparation

- Features (X) separated from target (y).
- Dataset split into:
    - 80% Training Data
    - 20% Testing Data

Algorithm Used

Linear Regression

Reason for selection:

- Simple and interpretable.
- Suitable baseline model for regression tasks.
- Fast training and prediction.

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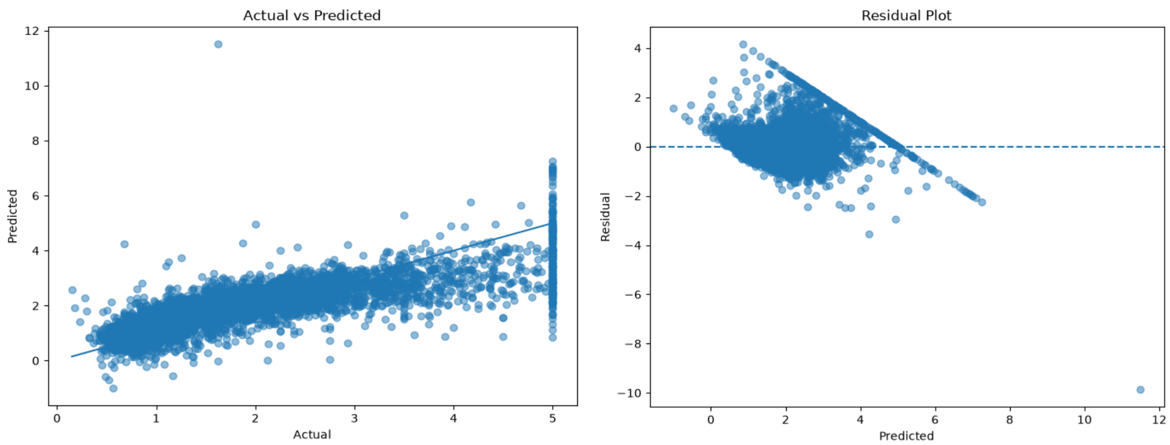
3. Model Metrics

Replace with your actual values.

Metric	Value
MAE	0.5332001304956555
RMSE	0.7455813830127763
R^2 Score	0.575787706032451

Interpretation

- **MAE** measures average prediction error.
- **RMSE** penalizes larger errors more heavily.
- **R^2 Score** indicates the proportion of variance explained by the model.



4. Key Findings

- Median Income is one of the strongest predictors of house value.
- Geographic location significantly impacts housing prices.
- Linear Regression provides reasonable prediction performance.
- The model can estimate house prices based on unseen inputs.

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5. Improvement Ideas

This section is explicitly requested, so make it its own heading.

Future Improvements

1. **Feature Engineering**
    - Create additional meaningful features.
    - Combine related variables.
2. **Outlier Handling**
    - Detect and remove extreme values.
    - Reduce model bias.
3. **Feature Scaling**
    - Standardize features before training.
4. **Advanced Models**
    - Random Forest Regressor
    - Gradient Boosting
    - XGBoost
5. **Hyperparameter Tuning**
    - Optimize model settings.
    - Improve prediction accuracy.

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6. Conclusion

A Linear Regression model was successfully trained on the California Housing dataset. The model achieved reasonable predictive performance and demonstrated the influence of income and geographical factors on housing prices. Future improvements using advanced regression algorithms and feature engineering may further increase prediction accuracy.

Submitted By:

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